

LOUIS-DAVID PIRON

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RESEARCH INTERESTS

My interests revolve around three questions about complex biological systems: how they evolve over time, at rest and under internal or external perturbations (dynamical modeling); which mechanisms drive their responses (causal inference); and how models can generalize to unseen interventions and guide which experiment to run next (active learning).

Keywords: generative modeling, data-driven dynamical system modeling, multimodal representation learning, causal inference, active learning.

EDUCATION

Université catholique de Louvain (UCLouvain) Louvain-la-Neuve, Belgium
M.Sc. in Data Science and Applied Statistics (ranked 1st/15) Sep '24 – Jun '26

- Graduated **Summa Cum Laude**. Focus: Deep Learning, Bioinformatics.

Université catholique de Louvain (UCLouvain) Louvain-la-Neuve, Belgium
B.Sc. in Business Engineering (ranked 12th/234) Sep '21 – Jun '24

- Graduated **Cum Laude**. Focus: Statistics, Econometrics, Supply Chain Optimization.

RESEARCH EXPERIENCE

Multimodal Molecular Representations for Virtual Cell modeling Jul '26 – Present
Advisors: Dr. J. Lequyer, Prof. J. Gootenberg *Abudayyeh–Gootenberg Lab, HMS*

- Investigating biologically informed molecular embeddings that capture drug effects beyond chemical structure, aligning several modalities using self-supervised learning, to help virtual cell models generalize to unseen drugs.
- Benchmarking structural and pretrained molecular representations as drug conditioning for single-cell perturbation models, on a ~40k-compound vocabulary curated from public perturbation atlases.

Continuous-Time Generative modeling of Glycemic Responses Sep '25 – Jun '26
Master's Thesis, Advisor: Prof. Laura Symul *UCLouvain*

- Associated thesis: [2]
- Introduced **CLODE**, a controlled Latent ODE-RNN in which meals act as exogenous inputs to continuous-time latent dynamics, opening the door to personalized trajectory forecasting and counterfactual simulation of dietary interventions.

Statistical Pre-processing of CE-LIF Glycomics Data Sep '25 – Apr '26
Scientific Assistant, with Prof. Laura Symul *SMCS, UCLouvain*

- Associated abstract: [1] (EuroBioC 2026)
- Developed the pre-processing and visualization modules of **glycomix**, an R package for CE-LIF glycomics: raw chromatogram extraction, denoising, baseline removal, and temporal alignment; wrote the vignettes and documentation.

Shrinkage Estimators for Forecast Combination Feb '26 – May '26
Research Intern, Advisor: Prof. Nathan Lassance *LFIN, UCLouvain*

- Conducted literature review and performed an empirical benchmark of forecast combination methods against a newly introduced shrinkage-based framework.

TEACHING

Teaching Assistantships 2025 – 2026
Teaching Assistant, exercise sessions *UCLouvain*

- **LINGE1222 | Multivariate Statistical Analysis**: introduced groups of 20 students to (un)supervised learning methods and programming in R. 2026
- **LIEPR1026 | Statistics**: introduced groups of 30 to 90 students to statistics and probability theory. 2025

PUBLICATIONS

Poster

- [1] M. Vast, B. Turner, K. M. Wheeler, O. Sokolovskaya, A. Grebin, **L.-D. Piron**, D. Kwon, K. Ribbeck, L. Symul. *glycomix: a package for pre-processing and extracting biological signal from CE-LIF glycomics data*. European Bioconductor Conference (**EuroBioC 2026**), Turku, Finland. Abstract presentation.

Thesis

- [2] **L.-D. Piron**. *Controlled Latent ODE-RNN for Continuous-Time Blood Glucose Modeling: Representation Learning, Factual and Counterfactual Predictions*. M.Sc. thesis, Université catholique de Louvain, 2026.

FELLOWSHIPS & AWARDS

- **Junior Orsi Fellowship** (\$4,700), Orsi Academy: one of 11 Belgian fellows selected from 250+ applicants for a funded research stay at Harvard Medical School 2026
- **Machine Learning Competition (LINFO2262)**, UCLouvain : ranked *2nd* on 179 Masters Students, the goal was to predict the toxicity of a drug.

INDUSTRY EXPERIENCE

PricewaterhouseCoopers (PwC)

Jun '25 – Sep '25

Data Scientist Intern, Deals & Analytics

Brussels, Belgium

- Conducted LLM-based sentiment analyses on web-scraped reviews and built an internal AI agent to automate client mapping.

LEADERSHIP & ENGAGEMENT

Expand Consulting (Junior Enterprise)

Mar '24 – Jul '25

Training Manager

Louvain-la-Neuve, Belgium

- Organized events on environmental, social and economic challenges reaching 300+ students; team awarded **Best Belgian Junior Enterprise ESG Approach 2025**.

Patro St. Gertrude de Carlsbourg

Sep '19 – Aug '23

Head of Staff & Activity Leader

Carlsbourg, Belgium

- Led 30 volunteers in a youth movement dedicated to civic education, coordinating camps for 100+ children.

École des Devoirs de Lauzelle

Sep '23 – Jun '26

Volunteer Tutor

Louvain-la-Neuve, Belgium

- Provided weekly homework support to primary and secondary school pupils.